A 1-MORPHISM OF FIBRED CATEGORIES IS CONTINUOUS AND COCONTINUOUS FOR THE INHERITED TOPOLOGIES

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Editorial note. No omission markers in-span; proof complete as written.

Allowed standard prerequisites (editorial).

- Continuous and cocontinuous functors. sites-definition-continuous, sites-definition-cocontinuous (cross-chapter, named). Fix the two conditions to be verified for F and the resulting morphism of topoi.
- Representable fibres go up. categories-lemma-fibred-category-representable-goes-up (cross-chapter, named). Identifies F(x_i x_x x') with F(x_i) x_{F(x)} F(x') in the continuity check.
- Morphism of topoi from cocontinuous functor. sites-lemma-cocontinuous-morphism-topoi, sites-lemma-pu-sheaf, sites-lemma-when-shriek (cross-chapter, named). Yield the final assertion f^{-1}(G)(x) = G(F(x)) from continuity and cocontinuity.

Source reference. Original reference 02XP: focused-seeds/geometry/S9-stk-topology-inherited-functorial/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 06N5: focused-seeds/geometry/S9-stk-topology-inherited-functorial/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00XJ: focused-seeds/geometry/S9-stk-topology-inherited-functorial/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00WV: focused-seeds/geometry/S9-stk-topology-inherited-functorial/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00XO: focused-seeds/geometry/S9-stk-topology-inherited-functorial/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the

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Source reference. Original reference 00XK: focused-seeds/geometry/S9-stk-topology-inherited-functorial/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00XR: focused-seeds/geometry/S9-stk-topology-inherited-functorial/ref-tags.txt, source lines 11–11. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let $F : \mathcal{X} \to \mathcal{Y}$ be a 1-morphism of fibred categories over $\mathcal{C}$. Then F is a continuous and cocontinuous functor between the structure of sites inherited from $\mathcal{C}$. Hence F induces a morphism of topoi $f : Sh(\mathcal{X}) \to Sh(\mathcal{Y})$ with $f_* = {}_sF = {}_pF$ and $f^{-1} = F^s = F^p$. In particular $f^{-1}(\mathcal{G})(x) = \mathcal{G}(F(x))$ for a sheaf $\mathcal{G}$ on $\mathcal{Y}$ and object x of $\mathcal{X}$.*

Proof. We first prove that F is continuous. Let $\{x_i \to x\}_{i\in I}$ be a covering of $\mathcal{X}$. By Categories, Definition 02XP the functor F transforms strongly cartesian morphisms into strongly cartesian morphisms, hence $\{F(x_i) \to F(x)\}_{i\in I}$ is a covering of $\mathcal{Y}$. This proves part (1) of Sites, Definition 00WV. Moreover, let $x' \to x$ be a morphism of $\mathcal{X}$. By Categories, Lemma 06N5 the fibre product $x_i \times_x x'$ exists and $x_i \times_x x' \to x'$ is strongly cartesian. Hence $F(x_i \times_x x') \to F(x')$ is strongly cartesian. By Categories, Lemma 06N5 applied to $\mathcal{Y}$ this means that $F(x_i \times_x x') = F(x_i) \times_{F(x)} F(x')$. This proves part (2) of Sites, Definition 00WV and we conclude that F is continuous.

Next we prove that F is cocontinuous. Let $x \in \mathrm{Ob}(\mathcal{X})$ and let $\{y_i \to F(x)\}_{i\in I}$ be a covering in $\mathcal{Y}$. Denote $\{U_i \to U\}_{i\in I}$ the corresponding covering of $\mathcal{C}$. For each i choose a strongly cartesian morphism $x_i \to x$ in $\mathcal{X}$ lying over $U_i \to U$. Then $F(x_i) \to F(x)$ and $y_i \to F(x)$ are both a strongly cartesian morphisms in $\mathcal{Y}$ lying over $U_i \to U$. Hence there exists a unique isomorphism $F(x_i) \to y_i$ in $\mathcal{Y}_{U_i}$ compatible with the maps to $F(x)$. Thus $\{x_i \to x\}_{i\in I}$ is a covering of $\mathcal{X}$ such that $\{F(x_i) \to F(x)\}_{i\in I}$ is isomorphic to $\{y_i \to F(x)\}_{i\in I}$. Hence F is cocontinuous, see Sites, Definition 00XJ.

The final assertion follows from the first two, see Sites, Lemmas 00XO, 00XK, and 00XR. □